

Analysis of Open-Address Hashing

Load Factor and Uniform Hashing

Let:

$$\alpha = \frac{n}{m}$$

where:

- n = number of stored elements
- m = number of slots

In open addressing:

$$n \leq m \quad \Rightarrow \quad \alpha \leq 1$$

We assume **uniform hashing**:

Each probe sequence

$$\langle h(k, 0), h(k, 1), \dots, h(k, m-1) \rangle$$

is equally likely to be any permutation of $\{0, 1, \dots, m-1\}$.

This is an idealized assumption used for analysis.

Unsuccessful Search

Theorem 1 (11.6). *Given an open-address hash table with load factor $\alpha < 1$, the expected number of probes in an unsuccessful search is at most*

$$\frac{1}{1 - \alpha}.$$

Proof

Let X be the number of probes in an unsuccessful search.

An unsuccessful search continues probing until it encounters an empty slot. Every probe except the last must hit an occupied slot.

Define event A_i :

A_i = event that the i th probe occurs and hits an occupied slot

Thus, for at least i probes to occur, the first $i - 1$ probes must all hit occupied slots.

$$\{X \geq i\} = A_1 \cap A_2 \cap \cdots \cap A_{i-1}$$

By the rule of conditional probability:

$$\Pr(X \geq i) = \Pr(A_1) \Pr(A_2|A_1) \cdots \Pr(A_{i-1}|A_1 \cdots A_{i-2})$$

Now we compute these probabilities.

Since there are n occupied slots among m total slots:

$$\Pr(A_1) = \frac{n}{m}$$

For $j > 1$, given that the first $j - 1$ probes hit distinct occupied slots, there remain:

$$n - (j - 1) \text{ occupied slots}$$

and

$$m - (j - 1) \text{ unexamined slots.}$$

Under uniform hashing, every remaining slot is equally likely to be probed next. Therefore,

$$\Pr(A_j|A_1 \cdots A_{j-1}) = \frac{n - (j - 1)}{m - (j - 1)}.$$

Since $n < m$, we have for all j :

$$\frac{n - (j - 1)}{m - (j - 1)} \leq \frac{n}{m} = \alpha.$$

Therefore,

$$\Pr(X \geq i) \leq \alpha^{i-1}.$$

Now we compute the expectation.

A useful identity for nonnegative integer random variables is:

$$E[X] = \sum_{i=1}^{\infty} \Pr(X \geq i).$$

Using the bound above:

$$E[X] \leq \sum_{i=1}^{\infty} \alpha^{i-1} = \sum_{i=0}^{\infty} \alpha^i.$$

This is a geometric series, and since $\alpha < 1$:

$$\sum_{i=0}^{\infty} \alpha^i = \frac{1}{1 - \alpha}.$$

Hence,

$$\boxed{E[X] \leq \frac{1}{1 - \alpha}}$$

□

Intuition Behind the Result

An unsuccessful search behaves like repeatedly “trying” slots until finding an empty one.

- We always perform the first probe.
- With probability approximately α , the first slot is occupied, so we probe again.
- With probability approximately α^2 , the first two slots are occupied.
- With probability approximately α^3 , the first three are occupied.

Thus, the expected number of probes behaves like:

$$1 + \alpha + \alpha^2 + \alpha^3 + \cdots = \frac{1}{1 - \alpha}.$$

Numerical Examples

If the table is half full:

$$\alpha = 0.5 \quad \Rightarrow \quad E[X] \leq \frac{1}{1 - 0.5} = 2.$$

If the table is 90% full:

$$\alpha = 0.9 \quad \Rightarrow \quad E[X] \leq \frac{1}{1 - 0.9} = 10.$$

This shows that performance deteriorates rapidly as α approaches 1.

Insertion

Corollary 1 (11.7). *Inserting an element requires at most*

$$\frac{1}{1 - \alpha}$$

probes on average.

Proof

Insertion consists of:

- Performing an unsuccessful search to find an empty slot.
- Placing the element in the first empty slot found.

Since the expected cost of an unsuccessful search is at most $\frac{1}{1-\alpha}$, the same bound applies to insertion.

$$E[\text{probes for insertion}] \leq \frac{1}{1 - \alpha}$$

□

Successful Search

Theorem 2 (11.8). *Given an open-address table with load factor $\alpha < 1$, the expected number of probes in a successful search is at most*

$$\frac{1}{\alpha} \ln \left(\frac{1}{1 - \alpha} \right).$$

Proof

A successful search for a key k follows exactly the same probe sequence that was used when k was inserted. This is because open addressing stores the key in the first empty slot encountered during insertion, and searching follows the identical probe sequence until it finds that key.

Suppose k was the $(i + 1)$ st key inserted into the table. At that time, exactly i keys were already present, so the load factor during its insertion was:

$$\frac{i}{m}.$$

By Theorem 11.6 (unsuccessful search bound), the expected number of probes needed to insert this key was at most:

$$\frac{1}{1 - i/m} = \frac{m}{m - i}.$$

Since a successful search for k retraces the same probes, the expected number of probes in searching for this key is also at most:

$$\frac{m}{m - i}.$$

Now we average over all n stored keys, assuming each key is equally likely to be searched for.

$$E[X] = \frac{1}{n} \sum_{i=0}^{n-1} \frac{m}{m - i}.$$

Factor out m :

$$= \frac{m}{n} \sum_{i=0}^{n-1} \frac{1}{m - i}.$$

Since $\alpha = n/m$, we have:

$$\frac{m}{n} = \frac{1}{\alpha}.$$

Thus,

$$E[X] = \frac{1}{\alpha} \sum_{i=0}^{n-1} \frac{1}{m - i}.$$

Reindex the summation by letting:

$$k = m - i.$$

As i goes from 0 to $n - 1$, k goes from m down to $m - n + 1$.

Thus:

$$E[X] = \frac{1}{\alpha} \sum_{k=m-n+1}^m \frac{1}{k}.$$

The sum

$$\sum_{k=m-n+1}^m \frac{1}{k}$$

is part of the harmonic series.

To bound it, we use the fact that for a decreasing function $1/x$:

$$\sum_{k=a}^b \frac{1}{k} \leq \int_{a-1}^b \frac{1}{x} dx.$$

Applying this inequality:

$$\sum_{k=m-n+1}^m \frac{1}{k} \leq \int_{m-n}^m \frac{1}{x} dx.$$

Compute the integral:

$$= \ln m - \ln(m - n) = \ln \left(\frac{m}{m - n} \right).$$

Since:

$$\alpha = \frac{n}{m} \quad \Rightarrow \quad m - n = m(1 - \alpha),$$

we obtain:

$$\frac{m}{m - n} = \frac{m}{m(1 - \alpha)} = \frac{1}{1 - \alpha}.$$

Therefore,

$$E[X] \leq \frac{1}{\alpha} \ln \left(\frac{1}{1 - \alpha} \right).$$

□

Intuition Behind the Result

When the table is nearly empty, most keys were inserted when few slots were occupied, so they required very few probes.

When the table becomes more full, later insertions require more probes.

A successful search averages over all these insertions, which leads to a harmonic-type growth. This is why the logarithm appears in the bound.

The growth is much slower than the unsuccessful case. Even when α is large, the performance remains moderate.

Numerical Examples

If $\alpha = 0.5$:

$$E[X] \leq \frac{1}{0.5} \ln \left(\frac{1}{0.5} \right) = 2 \ln 2 \approx 1.387.$$

If $\alpha = 0.9$:

$$E[X] \leq \frac{1}{0.9} \ln \left(\frac{1}{0.1} \right) \approx 2.559.$$

Notice that even at 90% load, successful search is still relatively small, while unsuccessful search would require about 10 probes.

Final Summary

Operation	Expected Probes
Unsuccessful Search	$\frac{1}{1 - \alpha}$
Insertion	$\frac{1}{1 - \alpha}$
Successful Search	$\frac{1}{\alpha} \ln \left(\frac{1}{1 - \alpha} \right)$

Open addressing performs well only when α is kept significantly below 1.